

3.1 环和域 (5)

定义(环中元素的幂) R 是环, ~~$r \in R$~~ 对 $\forall r \in R, \forall n \in \mathbb{Z}_{\geq 1}$, 定义

$$r^n := \underbrace{r \cdots r}_{n \uparrow r \text{ 相乘}}$$

$$r^0 := 1_R$$

Lemma: R 是环, 对 $\forall r \in R, \forall m, n \in \mathbb{Z}_{\geq 0}$, 有:

$$r^{m+n} = r^m r^n$$

Proof: 分4种情况讨论.

① $m \in \mathbb{Z}_{\geq 1}$ 且 $n \in \mathbb{Z}_{\geq 1}$. 则有:

$$\text{右边} = r^m r^n = (\underbrace{r \cdots r}_{m \uparrow r \text{ 相乘}}) \cdot (\underbrace{r \cdots r}_{n \uparrow r \text{ 相乘}}) = \underbrace{r \cdots r}_{(m+n) \uparrow r \text{ 相乘}} = r^{m+n} = \text{左边}.$$

↓
乘法结合律

② $m \in \mathbb{Z}_{\geq 1}$ 且 $n = 0$. 则有:

$$\text{右边} = r^m r^n = r^m r^0 = r^m \cdot 1_R = r^m = r^{m+0} = r^{m+n} = \text{左边}.$$

③ ~~$m \in \mathbb{Z}_{\geq 1}$~~ $m = 0$ 且 $n \in \mathbb{Z}_{\geq 1}$. 则有:

$$\text{右边} = r^m r^n = r^0 r^n = 1_R \cdot r^n = r^n = r^{0+n} = r^{m+n} = \text{左边}.$$

④ $m = 0$ 且 $n = 0$. 则有:

$$\text{右边} = r^m r^n = r^0 r^0 = 1_R \cdot 1_R = 1_R = r^0 = r^{0+0} = r^{m+n} = \text{左边}.$$



Lemma: R 是环, 对 $\forall r \in R, \forall m, n \in \mathbb{Z}_{\geq 0}$, 有:

$$(r^m)^n = r^{mn}$$

Proof: 分4种情况讨论.

① $m \in \mathbb{Z}_{\geq 1}$ 且 $n \in \mathbb{Z}_{\geq 1}$, 则有:

$$\text{左边} = (r^m)^n = \underbrace{r^m \cdot r^m \cdots r^m}_{n \text{ 个 } r^m \text{ 相乘}} = \underbrace{(\underbrace{r \cdots r}_{m \text{ 个 } r \text{ 相乘}}) \cdot (\underbrace{r \cdots r}_{m \text{ 个 } r \text{ 相乘}}) \cdots (\underbrace{r \cdots r}_{m \text{ 个 } r \text{ 相乘}})}_{n \text{ 个 "m 个 } r \text{ 相乘"}}$$

$$= \underbrace{r \cdots r}_{(n \times m) \text{ 个 } r \text{ 相乘}} = r^{mn} = \text{右边}.$$

↓
乘法结合律

② $m \in \mathbb{Z}_{\geq 1}$ 且 $n=0$. 则有:

$$\text{左边} = (r^m)^n = (r^m)^0 = |_R = r^0 = r^{m \times 0} = r^{mn} = \text{右边}.$$

③ $m=0$ 且 $n \in \mathbb{Z}_{\geq 1}$. 则有:

$$\text{左边} = (r^m)^n = (r^0)^n = |_R^n = \underbrace{|_R \cdots |_R}_{n \text{ 个 } |_R \text{ 相乘}} = |_R = r^0 = r^{mn} = \text{右边}$$

④ $m=0$ 且 $n=0$. 则有:

$$\text{左边} = (r^m)^n = (r^0)^0 = |_R^0 = |_R = r^0 = r^{mn} = \text{右边} \quad \square$$

Lemma: R 是环, ~~则有~~ 对 $\forall r \in R^\times$, 有:

$$(r^n)^{-1} = (r^{-1})^n \quad (\text{其中 } n \in \mathbb{Z}_{\geq 1})$$

Proof: 对 $\forall r \in R^\times$, 有:

$$\begin{aligned} \because R^\times \text{ 对取逆运算 } x \mapsto x^{-1} \text{ 封闭} & \quad \therefore r^{-1} \in R^\times \\ \because R^\times \text{ 对环 } R \text{ 的乘法运算封闭} & \quad \therefore r^n \in R^\times, (r^{-1})^n \in R^\times \quad \therefore (r^n)^{-1} \in R^\times \end{aligned}$$

$$\therefore r^n \cdot (r^{-1})^n = \underbrace{(r \cdots r)}_{n \text{ 个 } r \text{ 相乘}} \cdot \underbrace{(r^{-1} \cdots r^{-1})}_{n \text{ 个 } r^{-1} \text{ 相乘}} = \underbrace{(r \cdots r)}_{(n-1) \text{ 个 } r \text{ 相乘}} \cdot (r r^{-1}) \cdot \underbrace{(r^{-1} \cdots r^{-1})}_{(n-1) \text{ 个 } r^{-1} \text{ 相乘}}$$

$$= \underbrace{(r \cdots r)}_{(n-1) \text{ 个 } r \text{ 相乘}} \cdot \underbrace{(r^{-1} \cdots r^{-1})}_{(n-1) \text{ 个 } r^{-1} \text{ 相乘}} = \cdots = r r^{-1} = |_R$$

$$(r^{-1})^n \cdot r^n = \underbrace{(r^{-1} \dots r^{-1})}_{n \text{ 个 } r^{-1} \text{ 相乘}} \cdot \underbrace{(r \dots r)}_{n \text{ 个 } r \text{ 相乘}} = 1_R$$

$$\therefore r^n \cdot (r^{-1})^n = 1_R = (r^{-1})^n \cdot r^n$$

$$\therefore (r^n)^{-1} = (r^{-1})^n \quad \square$$

定义 (环中可逆元的负整数次幂). R 是环, 对 $\forall r \in R^\times$, ~~对~~ $\forall n \in \mathbb{Z}_{\geq 1}$,

$$\text{定义 } r^{-n} := (r^n)^{-1} = (r^{-1})^n.$$

Lemma: R 是环, 对 $\forall r \in R^\times$, ~~对~~ $\forall m, n \in \mathbb{Z}$, 有:

$$r^{m+n} = r^m r^n.$$

Proof: 分如下情况讨论:

	$m+n$	m	n
①	+	+	+
②	+	+	0
③	+	+	-
④	0	+	-
⑤	-	+	-
⑥	+	0	+
⑦	0	0	0
⑧	-	0	-
⑨	+	-	+
⑩	0	-	+
⑪	-	-	+
⑫	-	-	0
⑬	-	-	-

① ② ⑥ ⑦ 这4种情况已证.

③: 右边 $= r^m r^n = (r^{-1})^{-m} \cdot (r^{-1})^{-n} = (r^{-1})^{(-m)+(-n)} = (r^{-1})^{-(m+n)} = r^{m+n} = \text{左边}$
 $-m \in \mathbb{Z}_{\geq 1}, -n \in \mathbb{Z}_{\geq 1}$

③: 右边 $= r^m r^n = r^{(m+n)+(-n)} r^n = (r^{m+n} \cdot r^{-n}) \cdot r^n = r^{m+n} \cdot (r^{-n} \cdot r^n) = r^{m+n} \cdot (r^{-n} \cdot (r^{-n})^{-1})$
 $-n \in \mathbb{Z}_{\geq 1}$

$= r^{m+n} \cdot 1_R = r^{m+n} = \text{左边}$

④. 此时 $n = -m, m \in \mathbb{Z}_{\geq 1}$.

$\therefore \text{右边} = r^m r^n = r^m \cdot r^{-m} = r^m \cdot (r^m)^{-1} = 1_R = r^0 = r^{m+n} = \text{左边}$

⑤. 右边 $= r^m r^n = r^m \cdot r^{(m+n)+(-m)} = r^m \cdot (r^{m+n} \cdot r^{-m}) = r^m \cdot r^{(-m)+(m+n)}$

情况⑤已证.

$= r^m \cdot (r^{-m} \cdot r^{m+n}) = (r^m \cdot r^{-m}) \cdot r^{m+n} = (r^m \cdot (r^m)^{-1}) \cdot r^{m+n} = 1_R \cdot r^{m+n} = r^{m+n} = \text{左边}$

↓
情况③已证.

⑧, ⑫ 显然, ⑨ ⑩ ⑪ 的证明完全类似于 ③ ④ ⑤.



Lemma: R 是环, 对 $\forall r \in R^\times, \forall m, n \in \mathbb{Z}$, 有:

$$r^{mn} = (r^m)^n$$

Proof: 分如下情况讨论:

	mn	m	n
①	+	+	+
②	0	+	0
③	-	+	-
④	0	0	+
⑤	0	0	0
⑥	0	0	-
⑦	-	-	+
⑧	0	-	0
⑨	+	-	-

① ② ④ ⑤ 已证.

$$\textcircled{7}: \text{右边} = (r^m)^n = ((r^{-1})^{-m})^n = (((r^{-1})^{-m})^{-1})^{-n} = (((r^{-1})^{-m})^{-n})^{-1} = ((r^{-1})^{mn})^{-1} \\ = ((r^{mn})^{-1})^{-1} = r^{mn} = \text{左边}.$$

$$\textcircled{3}: \text{右边} = (r^m)^n = ((r^m)^{-n})^{-1} = (r^{-mn})^{-1} = r^{-(-mn)} = r^{mn} = \text{左边}$$

⑥ ⑧ 显然.

$$\textcircled{7}: \text{右边} = (r^m)^n = ((r^{-1})^{-m})^n = (r^{-1})^{-mn} = r^{-(-mn)} = r^{mn} = \text{左边}. \quad \square$$

Lemma: R 是环, $u \in R^x$, 对 $\forall r \in R$, 有:

$$\begin{array}{ccccc} r \in R^x & \Leftrightarrow & u \in R^x & \Leftrightarrow & ru \in R^x \\ \textcircled{1} & & \textcircled{2} & & \textcircled{3} \end{array}$$

Proof: $\textcircled{1} \Rightarrow \textcircled{2}$: $\because R^x$ 对环 R 的乘法成群, $u \in R^x, r \in R^x$

$$\therefore ur \in R^x, ru \in R^x.$$

$\therefore \textcircled{1} \Rightarrow \textcircled{2}$ 得证. $\textcircled{1} \Rightarrow \textcircled{3}$ 得证.

$$\textcircled{2} \Rightarrow \textcircled{1}: \because R^x \text{ 对取逆运算 } x \mapsto x^{-1} \text{ 封闭, } u \in R^x \therefore u^{-1} \in R^x.$$

$$\because R^x \text{ 对环 } R \text{ 的乘法成群, } u^{-1} \in R^x, ur \in R^x$$

$$\therefore u^{-1} \cdot (ur) \in R^x.$$

$$\therefore u^{-1} \cdot (ur) = (u^{-1}u) \cdot r = 1_R \cdot r = r \therefore r \in R^x$$

$\therefore \textcircled{2} \Rightarrow \textcircled{1}$ 得证.

$$\textcircled{3} \Rightarrow \textcircled{1}: \because R^x \text{ 对取逆运算 } x \mapsto x^{-1} \text{ 封闭, } u \in R^x \therefore u^{-1} \in R^x.$$

$$\because R^x \text{ 对环 } R \text{ 的乘法成群, } ru \in R^x, u^{-1} \in R^x$$

$$\therefore (ru)u^{-1} \in R^x$$

$$\therefore (ru)u^{-1} = r(uu^{-1}) = r \cdot 1_R = r \therefore r \in R^x \therefore \textcircled{3} \Rightarrow \textcircled{1} \text{ 得证. } \quad \square$$

定义(交换环) $(R, +, \cdot, 0_R, 1_R)$ 是环. 如果环 R 的乘法满足交换律:

$$\text{对 } \forall x, y \in R, \text{ 有: } xy = yx$$

则称 R 为交换环.

Lemma: R 是环. 则有: R 是交换环 $\Leftrightarrow Z(R) = R$

Proof: $(\Rightarrow)$: $\because Z(R) := \{z \in R : \text{对 } \forall x \in R, \text{ 有 } zx = xz\}$

$$\therefore Z(R) \subseteq R.$$

对 $\forall \lambda \in R$. $\because R$ 是交换环. $\therefore$ 对 $\forall x \in R$, 有 $\lambda x = x\lambda \quad \therefore \lambda \in Z(R)$

$$\therefore R \subseteq Z(R) \quad \therefore Z(R) = R$$

$(\Leftarrow)$: 对 $\forall x, y \in R$. 有:

$$\because y \in R = Z(R) \quad \therefore yx = xy \quad \because xy = yx \quad \therefore R \text{ 是交换环. } \square$$

定义(除环) $(R, +, \cdot, 0_R, 1_R)$ 是环. 如果 $R^\times = R \setminus \{0_R\}$, 则称环 R 是除环.

定义(域) $(R, +, \cdot, 0_R, 1_R)$ 是环. 如果环 R 是交换除环, 则称环 R 是域.

定义(子域) $(R, +, \cdot, 0_R, 1_R)$ 是域, $S \subseteq R$ 是域 R 的子环.

如果 $(S, +, \cdot, 0_R, 1_R)$ 是域, 则称 S 是 R 的子域.

Lemma: R 是环. 则有: $0_R \in R^\times \Leftrightarrow R = \{0_R\}$

Proof: $(\Rightarrow)$: $\because 0_R \in R^\times \quad \therefore \exists$ 唯一的 $0_R^{-1} \in R$, 使得 $0_R^{-1} \cdot 0_R = 1_R = 0_R \cdot 0_R^{-1}$

$$\therefore 1_R = 0_R \cdot 0_R^{-1} = 0_R \quad \therefore R = \{0_R\}.$$

$$(\Leftarrow): \because R = \{0_R\} \quad \therefore 1_R = 0_R \quad \therefore 0_R = 1_R \in R^\times \quad \square$$

定义(域中的除法). F 是域, 对 $\forall x \in F, \forall y \in F \setminus \{0_F\}$, 定义:

$$x/y = \frac{x}{y} := xy^{-1} = y^{-1}x$$

例: $(\mathbb{Z}, +, \cdot, 0, 1)$ 是环, $\mathbb{Z}^\times = \{\pm 1\}$

证明参见 Elementary-Number-Theory-note / 第一章——整除理论. 第1篇笔记.